

SUMMER INTERNSHIP REPORT

On

AI AUTOMATION AND FULL-STACK SOFTWARE DEVELOPMENT

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, NITISH KUMAR SINGH, hereby declare that the internship report titled “AI Automation and Full-Stack Software Development” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. The work presented in this report describes the projects and technical contributions undertaken during my Summer Internship with ScaleOn. References and project sources used for preparing this report have been acknowledged. I affirm that this report has not been submitted to any other institute for the same degree/diploma requirement.

Signature: _____

Student Name: NITISH KUMAR SINGH

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me during my Summer Internship. The internship provided me with an opportunity to work on practical software engineering problems involving AI automation, API integration, authentication, backend services and full-stack web development.

I am thankful to the ScaleOn team and mentors for providing a practical development environment and for their guidance throughout the assigned work. I also acknowledge the support of the School of Computer Science and Engineering, IILM University, Greater Noida, for making the internship an important part of my academic learning.

Finally, I express my gratitude to my parents, teachers, friends and everyone who encouraged me during the completion of this internship and report.

Signature: _____

Student Name: NITISH KUMAR SINGH

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INTERNSHIP COMPLETION CERTIFICATE

The original Internship Completion Certificate issued by the organization is to be attached on this page in the final submitted report.

**[ATTACH ORIGINAL INTERNSHIP COMPLETION CERTIFICATE
HERE]**

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4. PROJECT DESCRIPTION

4.1 Introduction

The Summer Internship focused on practical software development and AI-enabled automation. The work covered three related projects: a LinkedIn Integration Module for social-media automation, an AI Email Automation System for intelligent email classification and response generation, and the ScaleOn Study Portal, a full-stack internship management and learning platform. Together, these projects provided hands-on exposure to backend APIs, OAuth 2.0, AI services, frontend dashboards, databases, authentication, role-based access control and software architecture.

The project work was carried out in the context of the ScaleOn Internship Program 2026. The repositories used for this report document the implemented features, technology stacks, project structures and setup procedures.

4.2 Organization Profile

ScaleOn is the organization/context associated with the internship projects documented in the submitted repositories. The projects demonstrate work in software engineering, AI automation and development of internal/productivity platforms. The repositories identify the internship work as part of the ScaleOn Internship Program 2026 and include practical assignments and team-based development.

The internship environment involved working with modern development technologies such as Python, FastAPI, Streamlit, Google Gemini, LinkedIn APIs, Next.js, React, Node.js, Express.js, TypeScript, PostgreSQL and Prisma ORM.

4.3 Problem Statement

The internship addressed multiple real-world automation and platform-management problems. First, social-media publishing can require repeated manual authentication and posting actions. The LinkedIn Integration Module addresses this by providing a backend service that uses LinkedIn OAuth 2.0 and APIs to connect a user account and publish text or image posts.

Second, businesses receive large numbers of emails that require classification, prioritization and responses. The AI Email Automation System automates the initial response workflow by reading Gmail messages, classifying them using AI, generating context-aware replies and allowing a user to review, save or send those replies.

Third, internship teams need a centralized system for managing intern accounts, learning modules, assignments, sessions and progress. The ScaleOn Study Portal addresses this need through a full-stack application with separate admin and intern workflows, authentication, permissions, learning progress, assignments, support tickets and audit information.

4.4 Project Objectives

- Develop a secure LinkedIn integration using OAuth 2.0 without handling user passwords.
- Enable publishing of text and image posts through LinkedIn APIs.
- Build an AI-assisted email workflow that classifies messages by category, priority, intent and sentiment.
- Generate professional, context-aware email replies using Google Gemini.
- Allow users to review, edit, save as draft or send AI-generated email responses.
- Contribute to a full-stack Study Portal for managing interns, learning content, assignments and administrative workflows.
- Understand modern backend architecture, database integration, authentication and API-driven frontend development.
- Practice collaborative software development through GitHub-based project repositories.

4.5 Scope of the Project

The internship work covered three project areas. The LinkedIn module focused on authentication, profile retrieval, text/image publishing and account disconnection. LinkedIn analytics and article publishing were documented as unsupported under the standard developer access used by the project.

The AI Email Automation System covered inbox retrieval, AI classification, reply generation, Gmail draft creation, sending, search, organization and analytics. The project uses a lightweight CSV log and documents possible future migration to a relational database and multi-account support.

The Study Portal covered administrative and intern-facing functionality including JWT authentication, intern management, learning phases and modules, XP/streak tracking, leaderboard, assignments, live sessions, support tickets, role-based permissions, session management, login history and audit logs. The repository also contains architecture, database, API and setup documentation.

4.6 Technologies and Tools Used

Project	Major Technologies / Tools
LinkedIn Integration Module	Python, FastAPI, Uvicorn, httpx, SQLAlchemy, SQLite, LinkedIn OAuth 2.0, python-dotenv, python-multipart, Docker, HTML/JavaScript
AI Email Automation System	Python 3.10+, Streamlit, Plotly, Google Gemini API (gemini-2.5-flash), IMAP, SMTP, imaplib, smtplib, CSV logging
ScaleOn Study Portal	Next.js 16, React 19, TypeScript, Tailwind CSS 4, Node.js 22, Express.js, PostgreSQL, Prisma ORM, JWT, Framer Motion, Lucide React

4.7 System Architecture

The three projects follow different but complementary architectures. The LinkedIn Integration Module is a FastAPI backend organized into routers, service logic, OAuth handling, API client functions, database models and schemas. Its OAuth flow starts with a connect request, redirects the user to LinkedIn, receives the callback code, exchanges it for an access token, retrieves profile information, stores the connected account and then uses the saved token for posting.

The AI Email Automation System follows a pipeline architecture: Gmail inbox → email reader → duplicate check → Gemini classification → structured classification and draft reply → CSV logging → Streamlit dashboard → user review → save draft/send/delete/star actions. This separates email retrieval, AI reasoning, persistence and user interaction.

The Study Portal uses a full-stack client-server architecture. The frontend is implemented with Next.js/React and communicates with a Node.js/Express backend. PostgreSQL is accessed through Prisma ORM. Authentication uses JWT access and refresh tokens with httpOnly cookies, while middleware handles authentication, CSRF protection, rate limiting, validation and errors. The repository organizes backend functionality into modules, middleware, services and utilities.

LinkedIn Module	AI Email Automation	Study Portal
User → OAuth → FastAPI → LinkedIn API ↓ SQLAlchemy/SQLite	Gmail → IMAP → Gemini → CSV Log ↓ Streamlit → Review → Draft/SMTP	Next.js/React → Express/Node.js ↓ Prisma → PostgreSQL ↓ JWT/RBAC

4.8 Methodology

The development methodology followed an iterative engineering process. Requirements were first understood from the internship assignments and project documentation. The projects were then divided into modules and implemented incrementally. API integrations and authentication flows were tested using local development environments and demo interfaces. Environment variables were used for secrets instead of hardcoding credentials.

For the LinkedIn module, the main implementation sequence was OAuth connection, callback handling, token exchange, profile retrieval, account persistence and post publishing. For the email project, the sequence was email fetching, parsing, duplicate detection, AI classification, reply generation, logging and user-controlled sending/drafting. For the Study Portal, the work followed a modular full-stack approach involving frontend pages, backend modules, database models, authentication middleware and administrative/intern workflows.

Version control through GitHub supported code organization and collaboration. Project README files, architecture documentation, API documentation and setup guides were used to make the implementations reproducible.

4.9 Expected Outcomes

- Working backend integration capable of securely connecting a LinkedIn account and publishing supported content.
- Reduced manual effort in email triage through AI-based classification and reply generation.
- Human-in-the-loop email automation so that users retain control before sending responses.
- A centralized study and internship management platform with separate admin and intern workflows.
- Practical understanding of authentication, REST APIs, databases, AI APIs, frontend dashboards and deployment-oriented project structure.
- Improved software engineering skills in modular design, documentation, debugging, testing and GitHub collaboration.

4.10 Certificates of Completion and Communication Mail Proof

As required by the college format, the final report should include a copy of the internship completion certificate issued by the organization and relevant official communication/email proof. These documents were not included with the project

repository links provided for preparation of this report, so the original documents should be inserted before final submission.

Recommended attachments: (1) Internship Completion Certificate, (2) Internship offer/selection or joining communication if available, (3) official internship task/assignment communication if required, and (4) any mentor confirmation or completion communication available to the student.

PROJECT 1: LINKEDIN INTEGRATION MODULE

The LinkedIn Integration Module is a FastAPI backend service for a larger social-media automation tool. It connects a user account through LinkedIn OAuth 2.0, retrieves profile information and allows text and image posts to be published. The repository explicitly notes that scheduling/auto-posting at a future time is a separate module being built by another teammate.

- OAuth 2.0 login using LinkedIn's authorization flow.
- Profile fetching for connected users.
- Text-post publishing and image-post publishing.
- Disconnect functionality for removing saved account/token information.
- Automatic redirect back to the frontend after authentication.
- Dockerized deployment for consistent development environments.
- Demo HTML/JavaScript frontend for endpoint testing.

The codebase separates concerns into `router.py`, `service.py`, `oauth.py`, `client.py`, `models.py`, `schemas.py` and `exceptions.py`. The README documents endpoints for connect, callback, profile, post, image post and disconnect. It also documents that analytics and article endpoints return a not-supported response because those capabilities require partner-level access.

PROJECT 2: AI EMAIL AUTOMATION SYSTEM

The AI Email Automation System is an AI-powered workflow built for the ScaleOn Internship Program 2026 AI Agent Developer track. It reads Gmail messages, classifies them by category, priority, intent and sentiment, and uses Google Gemini to generate professional draft replies. The dashboard allows the user to review and edit the response before saving it as a Gmail draft or sending it.

- Fetch all emails or only unread messages.

- Classify emails into Sales, Support, HR, Finance, Marketing, Meeting, Orders and Spam categories.
- Determine priority, intent and sentiment.
- Generate context-aware AI replies using Gemini.
- Save replies to Gmail Drafts or send them through SMTP after review.
- Search emails by sender, subject, intent, category or content.
- Star/delete messages and maintain a lightweight processing log.
- Analytics dashboard for category and priority distribution.
- Duplicate-safe processing using message_id tracking.

The project uses Streamlit and Plotly for the dashboard, Python IMAP/SMTP libraries for email operations, Gemini for AI processing and CSV storage for the processing log. Future improvements documented in the repository include SQLite/PostgreSQL storage, Slack/WhatsApp notifications, Google OAuth2 login, public deployment and multi-account support.

PROJECT 3: SCALEON STUDY PORTAL

The ScaleOn Study Portal is a full-stack internship management and learning platform. The repository describes separate admin and intern workflows. Admins can manage intern accounts, batches, roles, permissions and learning content, while interns can track progress, complete modules, submit assignments and earn XP.

- Admin and intern authentication using JWT with rotating refresh tokens.
- Admin-created intern IDs and onboarding workflow.
- Intern CRUD operations, batch/role transfer and account activation/suspension.
- Database-driven role-based access control with live permission editing.
- Learning phases/modules with XP and streak tracking.
- Leaderboard, assignments, live sessions and support tickets.
- Session management, login history and audit logs.
- Admin assignment management, filtering, creation/editing/deletion and submission grading.

The repository uses Next.js 16, React 19, TypeScript and Tailwind CSS on the frontend, with Node.js 22 and Express.js on the backend. PostgreSQL is accessed through Prisma ORM. JWT authentication uses access and refresh tokens stored through httpOnly cookies. The codebase also includes middleware for authentication,

CSRF, rate limiting, validation and error handling, along with service and utility layers.

5. BIBLIOGRAPHY / REFERENCES

1. ScaleOn, “LinkedIn Integration Module,” GitHub repository, nitishchauhan002/linkedin-integration-module.
2. ScaleOn Internship Program 2026, “AI Email Automation System,” GitHub repository, nitishchauhan002/AI-Email-Automation-System.
3. ScaleOn, “ScaleOn Study Portal,” GitHub repository, amangovindrao/ScaleOn_Study_Portal.
4. LinkedIn Developer Platform documentation, OAuth 2.0 and LinkedIn API resources.
5. Google Gemini API / Google AI Studio documentation.
6. FastAPI, Streamlit, Next.js, React, Express.js, Prisma and PostgreSQL official documentation.